

TOPIC 32/60

How to Make LLMs Give Persistent Answers?

Controlling the Chaos.

LLMs are fundamentally random number generators. Here is how you lock down parameters to achieve deterministic, reliable outputs for enterprise software.

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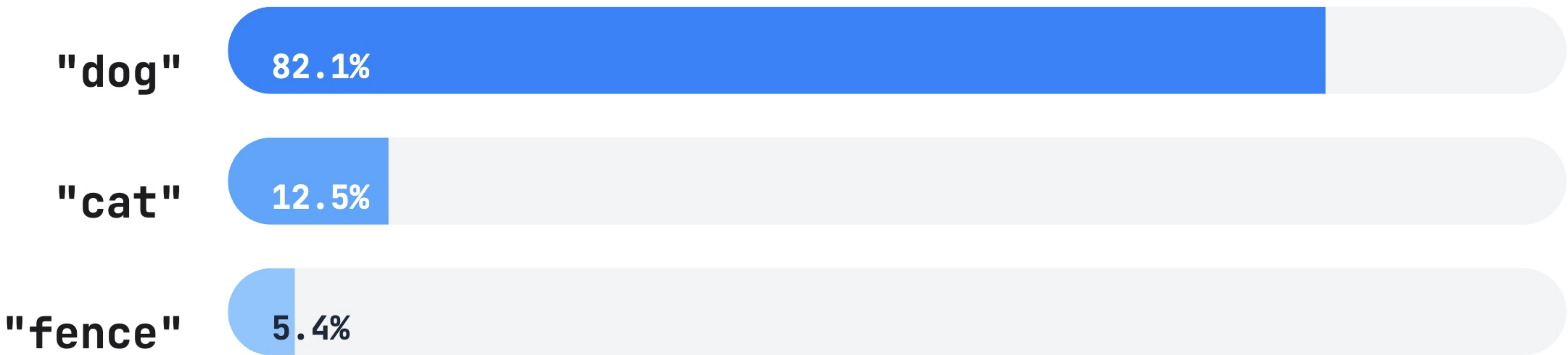
THE CORE PROBLEM

The Stochastic Brain

Language models do not "know" facts like a database. They are statistical engines that calculate the probability of the **next token**. By default, they roll a weighted die every time they speak.

INTERNAL PROBABILITY DISTRIBUTION

"The quick brown fox jumps over the lazy..."



By default, the LLM might randomly choose "cat" 12.5% of the time. We must stop it.



CONTROL LEVER 1

Temperature

Temperature controls the "creativity" or randomness of the response by flattening or sharpening the probability distribution before the die is rolled.

Temp = 0.0

GREEDY DECODING

The LLM acts like a robot. It will **always** pick the highest probability token. Use this for coding, JSON extraction, and facts.

Temp = 1.0+

HIGH ENTROPY

Flattens the probabilities, making lower-ranked words more likely to be chosen. Use this for poetry, brainstorming, and creative writing.

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CONTROL LEVER 2

Top-P (Nucleus)

Top-P dynamically trims the "long tail" of wildly incorrect words. It tells the model: *"Only pick from words whose combined probabilities add up to P."*

// Example: top_p = 0.90

1.	dog	82%	Sum: 82%
2.	cat	12%	Sum: 94% (Cutoff triggered)
3.	fence	5%	
4.	apple	1%	

* Result: The model is now physically barred from choosing "fence" or "apple", no matter what the random number generator does.



CONTROL LEVER 3

Top-K Sampling

While Top-P uses a percentage threshold, **Top-K uses a hard numerical limit**. It restricts the die roll to only the absolute Top \$K\$ most likely tokens.



Identical to Temperature = 0

Always picks the absolute #1 ranked word. 100% Deterministic.



Standard Safety Net

Sorts the entire 100,000 word vocabulary, grabs the top 50, and deletes the other 99,950 words before rolling the die.

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THE SECRET WEAPON

The `seed` Parameter

Due to GPU multithreading and floating-point math, even a Temperature of 0 can sometimes yield tiny variations. To achieve 100% reproducible scientific determinism, you must lock the PRNG (Pseudo-Random Number Generator).

API Request Payload:

```
{  
  "model": "gpt-4o",  
  "messages": [...],  
  "temperature": 0.0,  
  "seed": 42  
}
```

By providing a specific integer seed, the underlying mathematical random functions will execute in the exact same sequence every single time you call the API.

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PROMPT ENGINEERING

Few-Shot Anchoring

API parameters control the dice roll, but **Few-Shot Prompting controls the board**. By providing strict examples in the prompt, you mathematically skew the token probabilities toward your desired format.

SYSTEM PROMPT

User: Convert "Hello" to French.

Assistant: {"lang": "fr", "text": "Bonjour"}

User: Convert "Apple" to Spanish.

Assistant: {"lang": "es", "text": "Manzana"}

User: Convert "Dog" to German.

Assistant: *(The LLM is now anchored. It will output perfect JSON with near 100% certainty because the pattern probability outweighs raw text generation).*

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FORMAT ENFORCEMENT

Structured Outputs

Modern APIs allow you to physically lock the output syntax to a JSON Schema. The LLM's probability engine is overridden—if it tries to generate a token that breaks the JSON rules, the API instantly rejects it.

```
response_format = {  
  "type": "json_schema",  
  "schema": {  
    "name": "string",  
    "age": "number"  
  }  
}
```

Guaranteed Syntax

The model can no longer hallucinate introductory text like *"Sure, here is your JSON:"*. It is mathematically forced to begin with a `{` bracket.

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CHEAT SHEET

The Determinism Checklist



Set Temperature to 0.0

Stops creative sampling entirely.



Set a Static Seed (e.g., 42)

Forces consistent hardware PRNG state.



Provide 1-Shot or Few-Shot Examples

Anchors the probability space structurally.



Use Structured Outputs API

Forces strict schema adherence dynamically.

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Persistent Outputs

CONCEPT UNLOCKED



LLMs are probabilistic word calculators.



Temperature 0 forces greedy, robotic decoding.



Top-P & Top-K clip out the low-probability hallucinations.



Combine Seeds + Few-Shot for enterprise stability.

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